STAFF REVIEW · H446 021 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. A stack.

2. A queue.

3. The array index of another item.

4. Yes.

I Do — source worked steps

1. Make matrix — CODE:
int[,] graph = new int[3,3];

2. A—B — CODE:
graph[0,1] = 1;
graph[1,0] = 1;

3. B—C — CODE:
graph[1,2] = 1;
graph[2,1] = 1;

4. Read neighbours of B — CODE:
row 1 = 1 0 1
B connects to A and C

We Do — guided CHECK

1. Fill the same code
CODE:
graph[0,2] = 1;
graph[2,0] = 1;

2. Fill the same code
CODE:
graph[1,3] = ?;
CODE:
graph[1,3] = 1;

3. Fill the same code
CODE:
graph[0,2] = ?;
graph[2,0] = ?;
CODE:
graph[0,2] = 0;
graph[2,0] = 0;

4. Fill the same code
CODE:
if (graph[1,v] == ?)
Console.WriteLine(v);
CODE:
if (graph[1,v] == 1)
Console.WriteLine(v);

You Do — source mark schemes

1. For an undirected graph, every edge is recorded in both directions.
CODE:
int[,] graph = new int[4,4];
graph[0,1] = graph[1,0] = 1;
graph[0,2] = graph[2,0] = 1;
graph[2,3] = graph[3,2] = 1;
Matrix rows (0–3): 0110, 1000, 1001, 0010. Check that it is symmetrical across the diagonal.

2. Inspect row 0: a 1 marks a direct neighbour.
CODE:
for (int v = 0; v < 4; v++)
{
if (graph[0,v] == 1)
Console.WriteLine(v);
}
Output: 1 then 2. Vertex 3 is connected indirectly, not directly.

3. Adding an undirected edge sets two cells; removing one clears two.
CODE:
graph[1,3] = graph[3,1] = 1;
graph[0,2] = graph[2,0] = 0;
Edges afterwards: 0—1, 1—3, 2—3.

4. A directed edge is written in one matrix cell only.
CODE:
int[,] graph = new int[2,2];
graph[0,1] = 1;
Console.WriteLine(graph[0,1]); // 1
Console.WriteLine(graph[1,0]); // 0
0→1 exists; 1→0 does not. The matrix need not be symmetrical.

Exit Ticket — source answer

1. True. For an ordinary undirected edge between different vertices, the adjacency matrix records both directions.